

Rust Practice Set – Primitive & Compound Data Types

Solve each question without looking at the answer immediately. Try to write the code yourself, run it, and compare the output with the expected result. Topics covered: • Variables and mutability • Integer and floating-point types • Boolean and character types • Type annotations • Arrays and array indexing • Tuples and tuple destructuring • Slices • String slices (&str;) • String (heap-allocated strings) • Basic String methods

Question 1 – Variables and Mutability

Create a mutable variable `score` with value `10`. Add `15` to it and print the result.

Expected Output:

score = 25

Question 2 – Integer Types

Create variables using explicit types:

- `age → u8 = 20`
- `temperature → i16 = -12`

Print both values.

Expected Output:

Age: 20

Temperature: -12

Question 3 – Floating Point Numbers

Create two variables:

`price = 99.99 (f64)`

`discount = 10.5 (f32)`

Print both values.

Expected Output:

Price: 99.99

Discount: 10.5

Question 4 – Boolean and Character

Create:

```
is_rust_fun = true
```

```
grade = 'A'
```

Print both values.

Expected Output:

Rust is fun: true

Grade: A

Question 5 – Array Basics

Create an array named **numbers** containing five integers: 10, 20, 30, 40, 50.

Print the complete array using debug formatting.

Expected Output:

```
[10, 20, 30, 40, 50]
```

Question 6 – Array Indexing

Using the array from Question 5, print the first and last elements.

Expected Output:

First: 10

Last: 50

Question 7 – Array Length

Create an array of three cities: Mumbai, Delhi, Pune.

Print the length of the array.

Expected Output:

Total cities: 3

Question 8 – Tuples

Create a tuple named **student** containing:
("Abhishek", 20, true)
Print the entire tuple using debug formatting.

Expected Output:

("Abhishek", 20, true)

Question 9 – Tuple Destructuring

Using the tuple from Question 8, destructure it into separate variables and print each value.

Expected Output:

Name: Abhishek

Age: 20

Enrolled: true

Question 10 – String Slice (&str)

Create a string slice variable:

```
let language = "Rust";
```

Print the value.

Expected Output:

Language: Rust

Question 11 – String

Create a String using `String::from("Hello")`.

Append " Rust" using the appropriate String method and print it.

Expected Output:

Hello Rust

Question 12 – String Length

Create:

```
let name = String::from("Abhishek");
```

Print the number of bytes in the String.

Expected Output:

Length: 8

Question 13 – Slices

Create an array:

[5, 10, 15, 20, 25]

Create a slice containing 10, 15, and 20. Print the slice using debug formatting.

Expected Output:

[10, 15, 20]

Question 14 – String Slice from String

Create:

```
let text = String::from("Programming");
```

Create a slice containing only "Program" and print it.

Expected Output:

Program

Question 15 – Mixed Challenge

Create:

- A tuple: ("The Hobbit", 1937)
- A mutable String containing the author's name: "J.R.R. Tolkien"
- Append " (Author)" to the String.

Print the book title, year, and updated author name.

Expected Output:

Book: The Hobbit

Year: 1937

Author: J.R.R. Tolkien (Author)